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Python-for-machine-learning / 240901062-time series-3.ipynb

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315 lines (315 loc) · 7.77 KB

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In [2]:
```
import pandas as pd
import numpy as np
from datetime import datetime
dates=[datetime(2011,1,2),
          datetime(2011,1,5),
          datetime(2011,1,7),
          datetime(2011,1,8),
          datetime(2011,1,10),
          datetime(2011,1,12)]
ts=pd.Series(np.random.random(6),index=dates)
print("Date Index\n",ts)
```

```
Date Index
 2011-01-02    0.102535
2011-01-05    0.520488
2011-01-07    0.640825
2011-01-08    0.667942
2011-01-10    0.630108
2011-01-12    0.523548
dtype: float64
```

In [3]:
```
print("\nDate index alone from the series:\n",ts.index)
```

```
Date index alone from the series:
 DatetimeIndex(['2011-01-02', '2011-01-05', '2011-01-07', '2011-01-08',
               '2011-01-10', '2011-01-12'],
              dtype='datetime64[ns]', freq=None)
```

In [4]:
```
stamp=ts.index[2]
print("\nSecond index:\n",stamp)
```

```
Second index:
 2011-01-07 00:00:00
```

In [8]:
```
longer_ts=pd.Series(np.random.randn(1000),
                    index=pd.date_range("2000-01-01",periods=1000))
print("\nDate range upto 1000 days:\n",longer_ts)
```

```
Date range upto 1000 days:
 2000-01-01    0.234961
2000-01-02   -1.660426
2000-01-03   -0.704471
2000-01-04    0.654430
2000-01-05   -0.705605
                ...
2002-09-22   -0.453934
2002-09-23    1.768240
2002-09-24   -0.335533
2002-09-25    0.238938
2002-09-26    1.984772
Freq: D, Length: 1000, dtype: float64
```

In [11]:
```
print("\nDate range-2001 year:\n",longer_ts["2001"])
```

```
Date range-2001 year:
 2001-01-01    0.544858
2001-01-02    0.524951
2001-01-03    2.016821
2001-01-04   -0.856013
2001-01-05   -0.544740
                ...
2001-12-27    0.404264
2001-12-28   -0.407588
2001-12-29   -1.412428
2001-12-30    0.924385
2001-12-31    1.303876
Freq: D, Length: 365, dtype: float64
```

In [12]:
```
print("Date range=05 month 2001 year:\n",longer_ts["2001-05"])
```

```
Date range=05 month 2001 year:
 2001-05-01   -0.342538
2001-05-02    1.496571
2001-05-03    1.298218
2001-05-04    0.917625
2001-05-05    0.637372
2001-05-06    0.193743
2001-05-07   -1.238723
2001-05-08    1.545913
2001-05-09    0.159251
2001-05-10   -0.468467
2001-05-11   -0.300230
2001-05-12   -0.227435
2001-05-13   -1.750486
2001-05-14   -0.807199
2001-05-15   -1.182419
2001-05-16   -0.513105
2001-05-17    0.757962
2001-05-18   -0.167058
```

```
2001-05-19   -0.863502
2001-05-20   -0.084244
2001-05-21   -0.510563
2001-05-22   -0.676806
2001-05-23   -0.984481
2001-05-24    0.356051
2001-05-25   -0.387712
2001-05-26   -1.279625
2001-05-27   -0.767322
2001-05-28   -1.132376
2001-05-29   -0.787658
2001-05-30    2.225856
2001-05-31    0.264489
Freq: D, dtype: float64
```

In [13]:

```
print("\n Date range-first 15 dates in 02 month 2001 year:\n",longer_ts["2000-02"].head(15))
```

```
 Date range-first 15 dates in 02 month 2001 year:
 2000-02-01    0.410168
2000-02-02   -0.748233
2000-02-03    1.427570
2000-02-04    0.418073
2000-02-05   -0.184408
2000-02-06    0.161959
2000-02-07   -0.058343
2000-02-08    0.220545
2000-02-09    0.283928
2000-02-10    0.113561
2000-02-11   -0.519367
2000-02-12    1.854224
2000-02-13    0.382510
2000-02-14   -0.137414
2000-02-15   -1.194450
Freq: D, dtype: float64
```

In [14]:

```
print("\nDate range-from 26-09-2001\n",longer_ts[datetime(2000,9,26):])
```

```
Date range-from 26-09-2001
 2000-09-26   -0.202401
2000-09-27   -0.560996
2000-09-28   -0.349618
2000-09-29    1.222080
2000-09-30    0.557358
                ...
2002-09-22   -0.453934
2002-09-23    1.768240
2002-09-24   -0.335533
2002-09-25    0.238938
2002-09-26    1.984772
Freq: D, Length: 731, dtype: float64
```

In [16]:

```
print("\nDate   range   from    26-9-2001   to  25-9-2002:\n",longer_ts[datetime(2001,9,26):datetime(2002,9,25)])
```

```
Date   range   from    26-9-2001   to  25-9-2002:
 2001-09-26   -0.843767
2001-09-27   -0.751099
2001-09-28    0.434204
2001-09-29   -0.334272
2001-09-30    0.896862
                ...
2002-09-21   -1.184830
2002-09-22   -0.453934
2002-09-23    1.768240
2002-09-24   -0.335533
2002-09-25    0.238938
Freq: D, Length: 365, dtype: float64
```